

Rudra Patel

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EXPERIENCE

Research Assistant – NASA Collaborative Robotics & AI, ARCS Lab, CSUN *Feb 2025 – Present*

- Designed reinforcement learning-based navigation using ROS2 and PyTorch to improve autonomous rover pathfinding.
- Simulated Mars-like environments using GANs for robust testing of agent performance under varying terrains.
- Reduced localization error by 30% by optimizing spatial landmark detection and retrieval mechanisms.
- Collaborated with cross-functional teams to integrate diagnostic tooling and telemetry into firmware validation loops.

Data Scientist Intern – Indian Space Research Organization (ISRO) *Jan 2024 – May 2024*

- Built CNN-LSTM models in TensorFlow for rainfall prediction and mineral detection using satellite-based time-series data.
- Developed automated Python-SQL ETL pipelines to clean and preprocess over 35GB of multi-source sensor data.
- Created dashboards in Tableau and Power BI for visualizing prediction outcomes and model performance.
- Enabled 30% faster training cycles by optimizing data input pipelines and validation procedures.

Machine Learning Developer – Nxon AI Pvt. Ltd., Gujarat, India *May 2021 – Nov 2023*

- Built Retrieval-Augmented Generation (RAG) systems using LangChain, OpenAI, and FAISS to improve document search and answer generation.
- Deployed LSTM-based financial models with containerized batch pipelines for production-scale time-series inference.
- Designed modular Python scripts and CI/CD workflows to streamline development and testing in a multi-service environment.Reduced inference time by 50% through batch optimizations and query caching strategies.

PROJECTS

Legal Chatbot for Washington State Law

- Built a LLaMA-OpenAI RAG chatbot for answering law-based queries using semantically indexed legislation data.
- Integrated FAISS with LangChain for document vector search and implemented query filters for legal topic categorization.
- Helped reduce user research time by over 80% through accurate, source-linked response generation.

AI Health Assistant for Doctors and Patients

- Designed an LLM-driven symptom-to-treatment recommendation engine using domain-specific embedding models.
- Integrated explainability layers to provide interpretable AI responses based on clinical data trends.
- Increased recommendation accuracy by incorporating ranking models fine-tuned on patient query patterns.

Interview AI Agent

- Created an AI-driven mock interviewer using Whisper for voice input, Claude/OpenAI for question generation, and LangChain orchestration.
- Parsed resumes and job descriptions to generate role-aligned technical and behavioral questions in real time.
- Boosted user engagement by 60% through conversational flow improvements and dynamic voice-based follow-ups.

EDUCATION

California State University of Northridge, USA *Master of Computer Science, expected May 2026*

GPA: 4 , Relevant Coursework: Applied Machine Learning, Statistical Inference, Robotics, Data Visualization, NLP

CSPIT, Charusat University, India *Bachelor of Engineering in Information Technology, May 2024*

GPA: 8.7/10.0, 2nd rank among 28,000 engineering students

SKILLS AND EXPERTISE

- **Programming Languages:** Python, C++, SQL, Rust, GO, React, JAV, C#, Java, Javascript , R
- **Frameworks & Tools:** PyTorch, TensorFlow, ROS2, OpenCV, LangChain, Docker, Git, GitHub, Git, JAX, scikit-learn
- **Databases:** PostgreSQL, Spark, Hive,Airflow,Matplotlib, Tableau, Vector Databases (FAISS,Pinecone), RESTful APIs
- **Cloud Platforms:** AWS, Oracle Cloud Infrastructure, Microsoft Azure, GitHub Actions, CI/CD, Docker, Kusto
- **Core Concepts:**Generative AI, Retrieval-Augmented Generation (RAG), Query Optimization, Time Series Forecasting,Reinforcement Learning, Deep Learning, NLP, Statistical Modeling,Anomaly Detection, Model Validation

CERTIFICATIONS

- ORACLE CLOUD INFRASTRUCTURE 2024 GENERATIVE AI CERTIFIED PROFESSIONAL
- CAREER ESSENTIALS IN DATA ANALYSIS BY MICROSOFT AND LINKEDIN
- AZ-104 MICROSOFT AZURE ADMINISTRATOR BY KODEKLOUD

PUBLICATIONS

- GAN-based Image Super-Resolution: A Technical Perspective in *International Conference on ICT for Intelligent Systems (2023)*